

Zhenhao Liu

Undergraduate Researcher in Natural Language Processing

 [likc1606.github.io](https://github.com/LikC1606) ·  github.com/LikC1606 · [Westlake NLP Lab](#)

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RESEARCH FOCUS NLP / Language Modeling / Cognitive Alignment / Research Agents

EDUCATION

Westlake University

Undergraduate, Artificial Intelligence

Hangzhou, China

2024 cohort

RESEARCH EXPERIENCE

Westlake NLP Lab

Undergraduate Researcher; advised by Prof. Yue Zhang

Hangzhou, China

[Advisor profile](#)

- Lead research on data-efficient Chinese language modeling and cognitive alignment, studying how representation depth and corpus-derived features affect word- and sentence-level fMRI encoding.
- Design reproducible analyses with controlled shared-representation comparisons, participant-cluster bootstrap, leave-one analyses, and sensitivity checks to separate task-specific effects from selection uncertainty.

PUBLICATION

RepDuet at NLPCC 2026 Chinese BabyLM: Task-Specific Hybrid Representations for the Cognitive Track

[Model](#)

[Paper PDF](#)

Sole first author; accepted to NLPCC 2026 Shared Tasks Poster

- Developed RepDuet, a task-specific hybrid representation that combines intermediate contextual states with corpus statistics for word-level evaluation and upper-layer contextual states with lexical and confidence features for sentence-level evaluation.
- Ranked first in the eligible Cognitive Track after organizer reproduction and final-layer verification, achieving 0.33965 on the hidden evaluation; controlled development analyses found a positive margin over the strongest shared representation.

RESEARCH INTERESTS

- **Cognitive NLP and language modeling:** data-efficient Chinese language models, representation learning, and neural encoding.
- **Research agents:** language-model agents for literature search, information synthesis, and research workflows.
- **AI for science:** AI systems that support scientific discovery and the everyday work of researchers.

HONORS

2026 ICPC Asia East Continent Final

Silver Medal

2025 ICPC Asia Xi'an Regional

Gold Medal

2025 CCPC Zhengzhou Site

Gold Medal

2025 ICPC Asia Shenyang Regional

Silver Medal

TECHNICAL SKILLS

Programming Python, Git, command-line development, testing, and reproducibility

AI/NLP data-efficient language modeling, representation learning, cognitive NLP, LLM applications, language agents, evidence retrieval and synthesis, and AI for science

Engineering Flask, LangChain, REST APIs, Pytest, Ruff, Mypy, GitHub Actions, and LaTeX